

GAJENDRA DHANOLIYA

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[Codeforces](#)

[CodeChef](#)

EDUCATION

Indian Institute of Technology Delhi B.Tech in Electrical Engineering	2022 – 2026
Jawahar Navodaya Vidyalaya Sitapura, Bundi, Rajasthan CBSE Class XII	2020 – 2021
Jawahar Navodaya Vidyalaya Joura, Morena, Madhya Pradesh CBSE Class X	2018 – 2019

SCHOLASTIC ACHIEVEMENTS

- JEE Advanced:** Secured All India Rank in the top 0.5% out of 1M+ candidates nationwide 2022
- Competitive Programming:** Codeforces **Specialist** (1500+, [G.Dhanoliya](#)) and CodeChef **4 Star** (1800+, [gajenx7](#)); solved 1000+ problems across competitive programming platforms Present

EXPERIENCE

Software Engineer Intern ZeTheta Algorithms Pvt. Ltd.	Jun 2025 – Aug 2025
- Engineered an end-to-end automated eKYC platform handling Aadhaar-based identity verification, biometric authentication, and multi-step customer onboarding workflows, cutting manual onboarding time by 60% - Developed a Video KYC pipeline in OpenCV performing face matching against identity documents and passive liveness detection to block spoofing attempts, reaching 93% verification accuracy - Architected the verification backend — modular REST APIs, JWT-based authentication with role-based access control, and PostgreSQL-backed persistence — to support auditable, compliance-ready KYC workflows	

PROJECTS

RouteOS: Intelligent Logistics & Fleet Optimization Platform | *Python, FastAPI, OR-Tools, React, TypeScript, PostgreSQL/PostGIS, Redis, Docker* | [GitHub](#)

- Built a full-stack multi-vehicle route optimization platform solving the **capacitated VRP** with **time windows**, service durations, per-vehicle distance limits, and priority-weighted order assignment using **Google OR-Tools**
- Cut total fleet travel distance by **20–35%** against a nearest-neighbour baseline (**375 km vs. 576 km** on 100 orders across 12 vehicles) with **100% order-assignment** rates on **50–250** order benchmarks
- Streamed live vehicle simulation over **WebSockets** with traffic-disruption injection and **dynamic stop resequencing**, backed by **PostGIS** proximity queries and a **Redis**-cached analytics dashboard

Enterprise Document Intelligence Assistant | *Python, FastAPI, Celery, PostgreSQL + pgvector, Redis, Docker* | [GitHub](#)

- Built an **asynchronous document ingestion pipeline** handling **12+ formats** (PDF, DOCX, PPTX, XLSX, HTML, EML, images, ZIP) with **PyMuPDF** extraction, automatic **Tesseract OCR** fallback for scanned pages, and language detection
- Implemented **token-aware chunking** (`tiktoken`) across fixed-size, sentence-boundary, and markdown strategies with sliding overlap, storing **vector embeddings** in **pgvector** behind a pluggable OpenAI/HuggingFace provider layer
- Designed a **multi-tenant** FastAPI backend with **JWT** authentication, organization/project scoping, API-key and audit-log models, **async SQLAlchemy 2** with **Alembic** migrations, **Celery/Redis** background workers, and **Docker Compose** orchestration

Online Coding Judge | *Python, FastAPI, PostgreSQL, SQLAlchemy, Docker, JWT* | [GitHub](#)

- Built a competitive-programming judge backend with a **Docker-sandboxed execution engine** enforcing per-submission isolation — **network disabled**, **128 MB** memory cap, **50% CPU** quota, and a **2s** wall-clock timeout — to safely run untrusted user code
- Implemented an automated evaluation pipeline running submissions against sample and hidden test cases, returning **Accepted** / **Wrong Answer** / **Runtime Error** / **TLE** verdicts with execution-time and memory metrics
- Designed a **REST API** across **7 resource domains** (auth, problems, test cases, submissions, contests, contest-problems, dashboard) with **JWT + bcrypt** authentication, contest registration and score-ranked **leaderboards**, and per-user acceptance-rate analytics

TECHNICAL SKILLS

Languages: C++ (C++20), Python, TypeScript, SQL

Backend: FastAPI, REST API Design, WebSockets, Async I/O, Celery, SQLAlchemy 2, Pydantic, Alembic, JWT Authentication, RBAC, Multi-Tenant Architecture, Redis Caching

Databases: PostgreSQL, Redis, pgvector, PostGIS, Schema Design, Indexing, Query Optimization, Database Migrations

AI/ML: Embeddings, Vector Databases, Semantic Search, OCR (Tesseract, PyMuPDF), Computer Vision (OpenCV), Face Matching & Liveness Detection, Combinatorial Optimization (OR-Tools), Tokenization (tiktoken)

DevOps & Tools: Docker, Docker Compose, Git/GitHub, Linux, Prometheus, Structured Logging, Postman, LaTeX

Frontend: React, Tailwind CSS, TanStack Query, Leaflet, Recharts

Coursework: Operating Systems, DBMS, Computer Networks, Computer Architecture, Object-Oriented Programming, Probability & Stochastic Processes